Macroeconomics I — MPE Insper

Practice list: Measurement, Welfare and Growth — qualifier level (Listas 1–2)

Solutions and grading rubric

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Parameters are symbols with the stated admissible ranges; answers are closed forms, conditions, signs and limits.

Block 1 — Price indices, new goods, welfare and growth accounting

(Kurlat chs. 1–2, 5 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) A household with CES preferences (elasticity of substitution $\eta > 0$, $\eta \neq 1$) over two goods spends a share $\omega \in (0,1)$ of its budget on good 1 and $1-\omega$ on good 2 at base prices. The price of good 1 is multiplied by $R > 1$; the price of good 2 does not change. The log overstatement of the true change in the cost of living by the Laspeyres index L, $\ln(L/P^*)$ with P^* the exact index, is:

A. $\ln(\omega R + 1 - \omega) - \omega \ln R$

B. $\ln(\omega R + 1 - \omega) - \frac{1}{\eta - 1}\ln(\omega R^{\eta-1} + 1 - \omega)$

C. $\ln(\omega R + 1 - \omega) - \frac{1}{1-\eta}\ln(\omega R^{1-\eta} + 1 - \omega)$

D. $\ln(\omega R + 1 - \omega) - \ln(\omega R^{\eta-1} + 1 - \omega)$

Answer: C

Step — exact index. Under CES the cost of a util is $P = \left(\sum_i (p_i/b_i)^{1-\eta}\right)^{1/(1-\eta)}$ (Question 2(a); any $\eta > 0$, $\eta \neq 1$), with base shares $w_i^0 = (p_{i0}/b_i)^{1-\eta}/P_0^{1-\eta}$; so $P^* = \left(\sum_i w_i^0 R_i^{1-\eta}\right)^{1/(1-\eta)} = (\omega R^{1-\eta} + 1 - \omega)^{1/(1-\eta)}$, the power mean of order $1-\eta$ of $(R, 1)$ with weights $(\omega, 1-\omega)$.

Step — bias. $L = \sum_i w_i^0 R_i = \omega R + 1 - \omega$ is the power mean of order 1; logs give C. Power means rise with the order (Question 2(b)) and $1 - \eta < 1$, so the bias is positive and rises with η; it tends to 0 as $\eta \to 0$ (Leontief: Laspeyres is exact), and to second order in $\ln R$ it equals $(\eta/2)\,\mathrm{Var}(\ln R_i) = (\eta/2)\,\omega(1-\omega)(\ln R)^2$.

Step — distractors. A is the Cobb–Douglas index (the limit $\eta \to 1$) used for every η; B uses order $\eta - 1$, the current-weight formula of Question 2(b), with base weights; D makes two mistakes: order $\eta - 1$ with base weights, as B, and no root $1/(\eta - 1)$.

Ref: Kurlat (2020), ch. 1, §1.2 (base-year versus final-year prices); Jones (2020), ch. 2, §2.3.

b) (3 points) *Welfare × growth.* Over T years Brazil's mean consumption grows at g log points a year (initial mean one), the variance of log consumption (normal) goes from v_0 to v_T, and the survival probability of Kurlat's single-year experiment (life expectancy over 100) rises from a_0 to $a_T > a_0$; leisure is constant (absorbed into $\bar{u}$). Utility is Kurlat–Jones–Klenow with $\sigma = 1$ and positive flow utility at both dates. If λ scales all date-0 consumption so that Rawls is indifferent between date-0 and date-T Brazil, the average annual growth of consumption-equivalent welfare, $(\ln \lambda)/T$, is:

A. $g + \frac{v_0 - v_T}{2T} + \frac{1}{T}\left(\frac{a_T}{a_0} - 1\right)\left(\bar{u} + gT - \frac{v_T}{2}\right)$

B. $g + \frac{v_T - v_0}{2T} + \frac{1}{T}\left(\frac{a_T}{a_0} - 1\right)\left(\bar{u} + gT + \frac{v_T}{2}\right)$

C. $g + \frac{v_0 - v_T}{2T} + \frac{1}{T}\ln\frac{a_T}{a_0}$

D. $g + \frac{v_T - v_0}{2T} + \frac{1}{T}\left(1 - \frac{a_0}{a_T}\right)\left(\bar{u} + \frac{v_0}{2}\right)$

Answer: A

Step — expected utility. With $\sigma = 1$ a living person gets $\bar{u} + \ln c$; survival is independent of consumption, so Kurlat's (2.2.1) gives $\mathbb{E}[u] = a(\bar{u} + \mathbb{E}\ln c)$. If $\ln c \sim N(\mu, v)$, mean consumption is $e^{\mu + v/2}$, so $\mathbb{E}\ln c = \ln \bar{c} - v/2$: $\mathbb{E}\ln c_0 = -v_0/2$ and $\mathbb{E}\ln c_T = gT - v_T/2$.

Step — indifference. (2.2.2) with date 0 as the reference, $a_0(\bar{u}+\ln\lambda-v_0/2)=a_T(\bar{u}+gT-v_T/2)$, gives $\ln\lambda=(a_T/a_0)(\bar{u}+gT-v_T/2)-\bar{u}+v_0/2$. Adding and subtracting $\bar{u}+gT-v_T/2$ splits it into consumption, gT; inequality, $(v_0-v_T)/2$; and longevity, $(a_T/a_0-1)(\bar{u}+gT-v_T/2)>0$. Dividing by T gives A.

Step — distractors. B uses $\mathbb{E}\ln c=\ln\bar{c}+v/2$; C treats longevity as a proportional consumption gain, ignoring that a year is worth its flow utility; D makes two mistakes: it takes date T as the reference (scales date-T consumption), weighting date-0 flow utility by $1-a_0/a_T$, and uses $\mathbb{E}\ln c=\ln\bar{c}+v/2$, as B does.

Ref: Kurlat (2020), ch. 2, §2.2, eqs. (2.2.1)–(2.2.2) and "The Value of Life Expectancy".

c) (2 points, T or F) A Paasche price index can never exceed the corresponding Laspeyres index.
Answer: F

Step — Bortkiewicz decomposition. With base expenditure weights w_i, price relatives $R_i^p=p_{i1}/p_{i0}$ and quantity relatives $R_i^q=q_{i1}/q_{i0}$: $L=\sum_i w_iR_i^p\equiv\bar{R}^p$ and Paasche $=\sum_i p_{i1}q_{i1}/\sum_i p_{i0}q_{i1}=\sum_i w_iR_i^pR_i^q/\sum_i w_iR_i^q$, so Paasche$/L=1+\mathrm{Cov}_w(R^p,R^q)/(\bar{R}^p\bar{R}^q)$, with $\bar{R}^q\equiv\sum_i w_iR_i^q$.

Step — counterexample. Two goods, $w=(\frac{1}{2},\frac{1}{2})$, $R^p=(R,1)$ and $R^q=(Q,1)$ with $R,Q>1$ (demand shifts toward the good whose price rises): $\mathrm{Cov}_w=(R-1)(Q-1)/4>0$ and Paasche$/L=1+(R-1)(Q-1)/[(R+1)(Q+1)]>1$. With stable CES preferences Question 2(b) gives Paasche $\le P_1/P_0\le$ Laspeyres; the statement fails once tastes or qualities shift.

Ref: Kurlat (2020), ch. 1, §1.2; Jones (2020), ch. 2, §2.3.

d) (2 points, T or F) *PPP × productivity growth.* In Brazil and in the US labor is the only input and moves freely between tradables and non-tradables (linear technologies); consumers buy a Cobb–Douglas basket with the same non-tradable share $\gamma\in(0,1)$; tradables obey the law of one price; the exchange rate is constant. If Brazil's tradable productivity grows faster than the US's while non-tradable productivity grows equally in both, Brazil's price level relative to the US's rises, at a rate that increases with γ and would be zero if all goods were tradable.
Answer: T

Step — within a country. With $y_j=A_jL_j$, competition and one wage, $w=p_TA_T=p_NA_N$, so $p_N/p_T=A_T/A_N$. The unit cost of the basket is $P=\kappa_\gamma p_T^{1-\gamma}p_N^{\gamma}=\kappa_\gamma p_T(A_T/A_N)^{\gamma}$, with $\kappa_\gamma=(1-\gamma)^{\gamma-1}\gamma^{-\gamma}$.

Step — across countries. With $p_T^{BR}=Ep_T^{US}$, $P^{BR}/(EP^{US})=[(A_T^{BR}/A_N^{BR})/(A_T^{US}/A_N^{US})]^{\gamma}$, which grows at $\gamma[(g_T^{BR}-g_T^{US})-(g_N^{BR}-g_N^{US})]=\gamma\Delta$, $\Delta\equiv g_T^{BR}-g_T^{US}>0$: positive, increasing in γ and zero at $\gamma=0$: non-tradables get dearer, with weight γ.

Ref: Kurlat (2020), ch. 1, §1.2 (PPP comparisons); Romer (2012), ch. 4, §4.2 (Balassa–Samuelson effect).

Question 2. (15 points) Discursive. *Measurement × growth accounting.* Consumers have CES preferences over the set Ω_t of varieties available at date t,

$$U=\Big(\sum_{i\in\Omega_t}(b_ix_i)^{\frac{\eta-1}{\eta}}\Big)^{\frac{\eta}{\eta-1}},\qquad \eta>1,$$

where $b_i>0$ is a quality weight. The statistical agency (IBGE) observes prices and expenditure shares and builds its deflators from varieties present in both periods. Growth rates are in log points per period.

a) (3 points) Find the minimum expenditure needed to obtain one unit of utility at prices p, and each variety's expenditure share.

Step — problem. $\min\sum_i p_ix_i$ s.t. $\sum_i(b_ix_i)^{\rho}=1$, $\rho\equiv(\eta-1)/\eta\in(0,1)$. U is concave and homogeneous of degree one, so the upper contour set is convex and the first-order conditions give the minimum: $p_i=\mu\rho b_i^{\rho}x_i^{\rho-1}$, μ the multiplier.

Step — demands. $x_i=(\mu\rho)^{\eta}b_i^{\eta-1}p_i^{-\eta}$, using $1/(1-\rho)=\eta$ and $\rho/(1-\rho)=\eta-1$. The constraint gives $(\mu\rho)^{\eta-1}\sum_i(p_i/b_i)^{1-\eta}=1$.

Step — cost of a util. Spending $= (\mu\rho)^{\eta}\sum_i b_i^{\eta-1}p_i^{1-\eta} = (\mu\rho)^{\eta}(\mu\rho)^{1-\eta} = \mu\rho$, so

$$P = \Big(\sum_i (p_i/b_i)^{1-\eta}\Big)^{\frac{1}{1-\eta}}, \qquad w_i = \frac{p_i x_i}{P} = \frac{(p_i/b_i)^{1-\eta}}{P^{1-\eta}}.$$

By homotheticity P does not depend on utility: it is a true price index, and $\mu\rho = P$ is the marginal cost of a util.

Economic intuition. Only quality-adjusted prices p_i/b_i matter; shares fall with them, faster the larger η.

Ref: Kurlat (2020), ch. 1, §1.2 (substitution, quality and new-goods biases).

Grading rubric.

- FOC and demands: 1.5 pts; P and the shares: 1.5 pts. Formula quoted without derivation: max. 1 pt in the item.

b) (3 points) Keep the set of varieties fixed between dates 0 and 1. Express the exact index P_1/P_0 in terms of observable price relatives $R_i = p_{i1}/p_{i0}$ and shares, rank it against the Laspeyres and Paasche indices, and approximate the Laspeyres bias to second order in the dispersion of $\ln R_i$.

Step — base and current weights. By (a), $(P_1/P_0)^{1-\eta} = \sum_i (p_{i1}/b_i)^{1-\eta}/P_0^{1-\eta} = \sum_i w_i^0 R_i^{1-\eta}$, so $P_1/P_0 = M_{1-\eta}(R; w^0)$, where $M_r(R;w) \equiv \big(\sum_i w_i R_i^r\big)^{1/r}$ is the power mean of order r. Since $w_i^1 = w_i^0 R_i^{1-\eta}/\sum_j w_j^0 R_j^{1-\eta}$, $\sum_i w_i^1 R_i^{\eta-1} = 1/\sum_j w_j^0 R_j^{1-\eta}$, and $P_1/P_0 = M_{\eta-1}(R; w^1)$.

Step — ranking. Power means rise with the order: $d\ln M_r/dr = r^{-2}\sum_i \tilde{w}_i \ln(\tilde{w}_i/w_i) \geq 0$ with $\tilde{w}_i \equiv w_i R_i^r/\sum_j w_j R_j^r$, a Kullback–Leibler divergence, zero iff all R_i are equal. Laspeyres $= \sum_i w_i^0 R_i = M_1(R; w^0)$ and Paasche $= \big(\sum_i w_i^1/R_i\big)^{-1} = M_{-1}(R; w^1)$; since $1-\eta < 1$ and $\eta - 1 > -1$, Paasche $\leq P_1/P_0 \leq$ Laspeyres, with equality iff all R_i are equal.

Step — second order. With $z_i = \ln R_i$, mean $\bar{z}$ and variance V under w^0: $\sum_i w_i^0 e^{rz_i} = e^{r\bar{z}}(1 + r^2V/2 + O(|z-\bar{z}|^3))$, so $\ln M_r = \bar{z} + rV/2 + O(|z-\bar{z}|^3)$. Hence $\ln L - \ln(P_1/P_0) \approx V/2 - (1-\eta)V/2 = \eta V/2$.

Economic intuition. Laspeyres ignores substitution away from goods that became dearer; the gap grows with η and with price dispersion.

Ref: Kurlat (2020), ch. 1, §1.2; Jones (2020), ch. 2, §2.3.

Grading rubric.

- Exact index, both weightings: 1.5 pts; ranking proved: 1 pt (asserted: 0 for this line); second-order bias: 0.5 pt. Special case (two goods, a given η) used for the general one: max. 1.5 pts.

c) (3 points) Now some varieties are new at date 1 and none disappears. Find the exact index in terms of the index over continuing varieties and observable shares. What happens as $\eta \to \infty$ and as $\eta \to 1$?

Step — split the index. Let $\mathcal{C}$ be the continuing set, $P_t^{\mathcal{C}}$ the index of (a) computed over $\mathcal{C}$ only, and $\lambda_t \equiv \sum_{i\in\mathcal{C}}(p_{it}/b_i)^{1-\eta}/\sum_{i\in\Omega_t}(p_{it}/b_i)^{1-\eta}$, which by (a) is the expenditure share of continuing varieties at t ($\lambda_0 = 1$). Then $P_t^{1-\eta} = (P_t^{\mathcal{C}})^{1-\eta}/\lambda_t$.

Step — formula. Raising to $1/(1-\eta)$: $P_t = P_t^{\mathcal{C}}\lambda_t^{1/(\eta-1)}$, so $P_1/P_0 = (P_1^{\mathcal{C}}/P_0^{\mathcal{C}})\,(\lambda_1/\lambda_0)^{1/(\eta-1)}$, which needs only shares, not the qualities b_i of the new varieties.

Step — limits. With $\lambda_1 < 1$ the log of the correction is $\ln\lambda_1/(\eta-1) < 0$. It tends to 0 as $\eta \to \infty$ (perfect substitutes add nothing beyond their price) and to $-\infty$ as $\eta \to 1^+$, so $P_1/P_0 \to 0$: new varieties become infinitely valuable.

Economic intuition. New varieties raise what a unit of spending buys; their share reveals by how much, and η turns it into a price.

Ref: Kurlat (2020), ch. 1, §1.2 (new-goods bias).

Grading rubric.

- λ_t as an observable share: 1 pt; formula derived: 1.5 pts; both limits, argued: 0.5 pt (asserted: 0 for this line).

d) (3 points) The agency measures the continuing-variety consumption index exactly. New varieties appear only among consumption goods, take a share $\nu \in (0,1)$ of date-1 consumption spending, and none disappears. Investment goods (nominal GDP share $s_I \in (0,1)$, $s_C = 1 - s_I$) have their own deflator bias $\beta_I \in [0, g_K + \delta)$ from unmeasured quality change (g_K: true capital growth; $\delta > 0$: depreciation); labor and the capital share $\alpha \in (0,1)$ are measured correctly. On a balanced growth path, find the consumption-deflator bias β_C, show that measured capital growth converges and at what speed, and find the distortion $\Delta \equiv g_A^m - g_A$ of the measured Solow residual; compare $\beta_I = 0$ with $\beta_I = \beta_C$.

Step — consumption bias. By (c) with $\lambda_0 = 1$ and $\lambda_1 = 1 - \nu$, true inflation is $\ln(P_1/P_0) = \ln(P_1^{\mathcal{C}}/P_0^{\mathcal{C}}) + \ln(1-\nu)/(\eta-1)$, so the measured deflator overstates it by $\beta_C = -\ln(1-\nu)/(\eta-1) > 0$.

Step — measured output. Real GDP growth is the share-weighted growth of its components, each equal to nominal growth minus its deflator: $g_Y^m = s_C(g_C - \beta_C) + s_I(g_I - \beta_I) = g_Y - (s_C\beta_C + s_I\beta_I)$.

Step — measured capital. Perpetual inventory (continuous time): $\dot{K}^m = I^m - \delta K^m$, with I^m growing at $g_K - \beta_I$ on the balanced path, so $x \equiv K^m/I^m$ obeys $\dot{x} = 1 - (g_K - \beta_I + \delta)x$. As $g_K - \beta_I + \delta > 0$, x converges monotonically to $1/(g_K - \beta_I + \delta)$ at that rate (half-life $\ln 2/(g_K - \beta_I + \delta)$), and $\dot{K}^m/K^m = 1/x - \delta \to g_K - \beta_I$: understated by β_I.

Step — residual. By Kurlat's (5.4.2), $g_A^m = g_Y^m - \alpha g_K^m - (1-\alpha)g_L$ and $g_A = g_Y - \alpha g_K - (1-\alpha)g_L$, so

$$\Delta \equiv g_A^m - g_A = -(s_C\beta_C + s_I\beta_I) + \alpha\beta_I = -s_C\beta_C - (s_I - \alpha)\beta_I.$$

Step — two cases. $\beta_I = 0$: $\Delta = -s_C\beta_C$. $\beta_I = \beta_C$: $\Delta = -(1-\alpha)\beta_C$, using $s_C + s_I = 1$. Both are negative, and $\Delta(\beta_C) - \Delta(0) = (\alpha - s_I)\beta_C$: the uniform bias distorts less iff $\alpha > s_I$.

Economic intuition. Understated output lowers measured TFP growth and understated capital raises it; equipment quality change is embodied technical progress (Romer's problem 1.12), booked as TFP.

Ref: Kurlat (2020), ch. 5, §5.4, eq. (5.4.2); Romer (2012), ch. 1, §1.7 and problem 1.12.

Grading rubric.

- β_C: 0.5 pt; output bias: 0.5 pt; capital (convergence, speed, bias β_I): 1 pt, convergence asserted: 0 for this line; residual and the two cases: 1 pt. Caps: capital channel ignored: max. 1.5 pts; $\beta_I = \beta_C$ used as the general case: max. 1.5 pts.

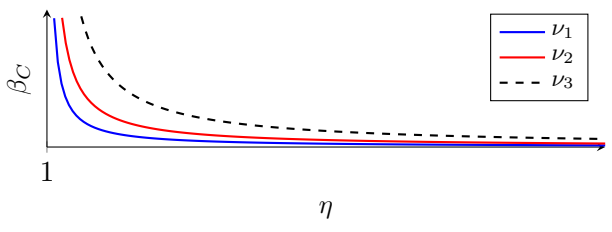
e) (3 points) *Parameter analysis.* How do β_C and Δ depend on η, ν, β_I, α and s_I? Prove and tabulate the signs. Find the β_I^* at which Δ changes sign, when it exists, and how it moves with the parameters. Give the limits of β_C as $\eta \to 1^+$, $\eta \to \infty$ and $\nu \to 0$, and draw β_C against η for three new-variety shares $\nu_1 < \nu_2 < \nu_3$.

Step — signs of β_C. From $\beta_C = \ell/(\eta-1)$ with $\ell \equiv -\ln(1-\nu) > 0$: $\partial_\eta \beta_C = -\ell/(\eta-1)^2 < 0$, $\partial_\nu \beta_C = 1/[(1-\nu)(\eta-1)] > 0$ and $\partial_\eta^2 \beta_C = 2\ell/(\eta-1)^3 > 0$; β_C does not depend on β_I, α or s_I.

Step — signs of Δ. From $\Delta = -(1-s_I)\beta_C - (s_I - \alpha)\beta_I$: $\partial_\eta \Delta = -(1-s_I)\partial_\eta\beta_C > 0$, $\partial_\nu \Delta = -(1-s_I)\partial_\nu\beta_C < 0$, $\partial_{\beta_I}\Delta = \alpha - s_I$, $\partial_\alpha \Delta = \beta_I \geq 0$ and $\partial_{s_I}\Delta = \beta_C - \beta_I$.

Step — sign switch. Δ is affine in β_I with intercept $-s_C\beta_C < 0$. If $\alpha > s_I$ it has the unique root $\beta_I^* = s_C\beta_C/(\alpha - s_I) > 0$, and $\Delta > 0$ iff $\beta_I > \beta_I^*$ (admissible only if $\beta_I^* < g_K + \delta$); if $\alpha \leq s_I$, $\Delta < 0$ for every $\beta_I \geq 0$. Moreover $\partial_\alpha \beta_I^* = -s_C\beta_C/(\alpha - s_I)^2 < 0$, $\partial_{s_I}\beta_I^* = (1-\alpha)\beta_C/(\alpha - s_I)^2 > 0$, and $\beta_I^* \propto \beta_C$ inherits the signs of β_C in η and ν. Table: an expression stands for its sign; row β_I^* assumes $\alpha > s_I$.

	η	ν	β_I	α	s_I
β_C	$-$	$+$	0	0	0
Δ	$+$	$-$	$\alpha - s_I$	β_I	$\beta_C - \beta_I$
β_I^*	$-$	$+$	0	$-$	$+$



Step — limits and reading. $\beta_C \to \infty$ as $\eta \to 1^+$ (new varieties become essential), $\beta_C \to 0$ as $\eta \to \infty$, and $\beta_C = \nu/(\eta-1) + O(\nu^2) \to 0$ as $\nu \to 0$: the curves fall, are convex, ordered

by ν, asymptotic to $\eta = 1$ and to the axis. Knife-edge $\alpha = s_I$: $\Delta = -s_C\beta_C$ for all β_I, and $\beta_I^* \to \infty$ as $\alpha \downarrow s_I$. Reading: on a one-good balanced path $I = (g_K + \delta)K$ and $\alpha = f'(k)K/Y$, so $\alpha - s_I = (K/Y)(f'(k) - \delta - g_K)$: $\alpha > s_I$ iff capital is below the golden rule (Kurlat (4.3.4)), the dynamically efficient case, where faster equipment quality change shrinks the TFP understatement and reverses it above β_I^*. The consumption bias is large for differentiated goods with many new varieties, not for commodities.

Ref: Kurlat (2020), ch. 1, §1.2; ch. 4, §4.3, eq. (4.3.4) (golden rule); ch. 5, §5.4.

Grading rubric.

- Signs proved: 1 pt; table: 0.5 pt; β_I^* (condition, signs): 0.5 pt; limits and reading: 0.5 pt (a limit asserted: 0 for this line); signs without proof or reading: max. 1.5 pts. Graph, scored separately: labeled axes, three decreasing convex curves ordered by ν, asymptote $\eta = 1$: 0.5 pt.

Block 2 — Growth, capital flows and national income

(Kurlat chs. 4–5; Romer chs. 1, 4 • 25 points)

Question 3. (10 points) Objective items.

a) (3 points) *Development accounting × capital flows.* Let $Y = AK^\alpha(hL)^{1-\alpha}$, $\alpha \in (0,1)$, with the same A in Brazil and the US. Brazil's output per worker is a fraction ϕ of the US's and its human capital per worker a fraction χ, with $0 < \phi < \chi < 1$. The ratio of marginal products of capital (Brazil/US) implied by the closed-economy model is:

A. $(1/\phi)^{(1-\alpha)/\alpha}$

B. $(\chi/\phi)^{1-\alpha}$

C. $(1/\phi)^{1-\alpha}$

D. $(\chi/\phi)^{(1-\alpha)/\alpha}$

Answer: D

Step — MPK in observables (production function). Per worker $y = Ak^\alpha h^{1-\alpha}$, $k = K/L$, and $MPK = \partial Y/\partial K = \alpha y/k$. Inverting, $k = \left(y/(Ah^{1-\alpha})\right)^{1/\alpha}$, so $MPK = \alpha A^{1/\alpha}(y/h)^{-(1-\alpha)/\alpha}$.

Step — ratio and distractors. With common A, α: $MPK_{BR}/MPK_{US} = (\chi/\phi)^{(1-\alpha)/\alpha} > 1$, falling in α (log-derivative $-\ln(\chi/\phi)/\alpha^2$) and tending to 1 as $\chi \to \phi$. A sets $\chi = 1$ (no human capital); B puts y/h for k/h in $MPK \propto (k/h)^{\alpha-1}$; C makes both mistakes. All four exceed 1; A exceeds C and D, which exceed B, as $\chi < 1$ and $(1-\alpha)/\alpha > 1 - \alpha$.

Economic intuition. Capital should flow to Brazil; that it does not (Lucas paradox) points to a lower A (see 4(a)).

Ref: Kurlat (2020), §5.3 (Lucas, 1990); Romer (2012), §1.6, eq. (1.32).

b) (3 points) *Growth × welfare.* Output per effective worker is $f(k) = A[\alpha k^\psi + 1 - \alpha]^{1/\psi}$ with $A > 0$, $\alpha \in (0,1)$, $\psi < 1$, $\psi \neq 0$ (elasticity of substitution $\sigma = 1/(1-\psi)$), and $\dot{k} = sf(k) - Nk$ with $N \equiv n + g + \delta > 0$. When a Golden-Rule capital stock exists, the saving rate that attains it is:

A. $(\alpha A^\psi)^{1-\psi} N^\psi$

B. $(\alpha A^\psi)^{-1/(1-\psi)} N^{\psi/(1-\psi)}$

C. $(\alpha A^\psi)^{1/(1-\psi)} N^{-\psi/(1-\psi)}$

D. $(\alpha A^\psi)^{-(1-\psi)} N^{-\psi}$

Answer: C

Step — Golden Rule and f'. In a steady state $sf(k) = Nk$, so $c = f(k) - Nk$, strictly concave, peaks where $f'(k_{GR}) = N$. As $[\alpha k^\psi + 1 - \alpha]^{1/\psi} = f/A$: $f'(k) = \alpha A[\alpha k^\psi + 1 - \alpha]^{(1-\psi)/\psi} k^{\psi-1} = \alpha A^\psi (f/k)^{1-\psi}$.

Step — saving rate. $f' = N$ gives $f/k = (N/\alpha A^{\psi})^{1/(1-\psi)}$; the steady state gives $s_{GR} = Nk/f$. Hence

$$s_{GR} = (\alpha A^{\psi})^{\frac{1}{1-\psi}} N^{-\frac{\psi}{1-\psi}} = \alpha^{\sigma} \Big(\frac{A}{N}\Big)^{\sigma-1} = \frac{k_{GR}\, f'(k_{GR})}{f(k_{GR})},$$

the capital share at k_{GR}, equal to α only in the limit $\sigma \to 1$ (Cobb–Douglas).

Step — existence. $f/k = A[\alpha + (1-\alpha)k^{-\psi}]^{1/\psi}$ strictly decreases, so f' falls from ∞ to $A\alpha^{1/\psi}$ ($\psi > 0$) or from $A\alpha^{1/\psi}$ to 0 ($\psi < 0$): k_{GR} exists iff $N > A\alpha^{1/\psi}$ ($\psi > 0$) or $N < A\alpha^{1/\psi}$ ($\psi < 0$), both $\iff \alpha(A/N)^{\psi} < 1 \iff s_{GR} < 1$. Otherwise c rises without bound ($\psi > 0$) or no $s < 1$ sustains any $k > 0$ ($\psi < 0$).

Step — distractors. A raises to $1-\psi$ instead of $1/(1-\psi)$ when solving $f' = N$; B inverts $sf = Nk$ (it is $1/s_{GR}$); D makes both mistakes. As $\psi \to 0$, A and C tend to α and B, D to $1/\alpha$: the Cobb–Douglas check cannot separate A from C.

Economic intuition. The Golden Rule reinvests capital's income; as $\partial \ln s_{GR}/\partial \ln(A/N) = \sigma - 1$, only Cobb–Douglas gives "save α".

Ref: Kurlat (2020), §4.3; Romer (2012), §1.4 (golden rule) and §1.8 (elasticity of substitution).

c) (2 points, T or F) Let $Y = [\alpha(BK)^{\psi} + (1-\alpha)(AL)^{\psi}]^{1/\psi}$, $\psi < 1$, $\psi \neq 0$, where B, A, L grow at the constant rates $g_B \geq 0$, g, n. A path on which K and Y grow at a common constant rate exists only if $g_B = 0$: technical progress must be purely labor-augmenting.
Answer: T

Step — proof. $(Y/K)^{\psi} = u + v$, $u \equiv \alpha B^{\psi}$, $v \equiv (1-\alpha)(AL/K)^{\psi}$; on such a path $u, v > 0$ grow at constant rates ψg_B, ψg_x ($g_x = g + n - g_K$) and $\dot{u} + \dot{v} = \psi(g_B u + g_x v) = 0$. If $g_B \neq 0$, $g_x = -g_B u/v$ has the sign opposite to g_B, yet a constant u/v needs $g_x = g_B$: contradiction. So $g_B = 0$, then $g_x = 0$: K grows at $n + g$.

Step — the exception. Cobb–Douglas ($\psi \to 0$): $(BK)^{\alpha}(AL)^{1-\alpha} = K^{\alpha}(B^{\alpha/(1-\alpha)}AL)^{1-\alpha}$, so it binds only if $\sigma \neq 1$.

Economic intuition. With $\sigma \neq 1$, capital-augmenting progress moves factor shares forever.

Ref: Romer (2012), §1.2 (footnote on Cobb–Douglas) and problem 1.12.

d) (2 points, T or F) *Growth × open economy.* In a small open Solow economy with neoclassical $Y = F(K, AL)$, perfect capital mobility at the world rate r^* and residents who save a constant fraction s of national income, a permanent rise in s permanently raises GDP per worker.
Answer: F

Step — proof. Firms rent capital until $f'(k) = r^* + \delta$, $k = K/(AL)$; f' strictly decreases, so $k^* = f'^{-1}(r^* + \delta)$ is unique and free of s: GDP per worker is $Af(k^*)$ at every date. Saving changes who owns capital (net foreign assets, GNI: Question 4). The claim holds in a closed economy and for GNI per worker: that is the trap.

Economic intuition. Capital flows, not domestic saving, finance the domestic capital stock.

Ref: Romer (2012), problem 4.4; Kurlat (2020), §1.1 (GDP measures production inside the country).

Question 4. (15 points) Discursive. *Growth × national accounts × capital flows.* Brazil as a small open economy. Output is $Y = K^{\alpha}(AhL)^{1-\alpha}$, $\alpha \in (0,1)$; capital moves freely and earns the world rate $r^* > 0$ net of depreciation $\delta > 0$; A grows at $g \geq 0$ and L at $n \geq 0$, with $n + g > 0$; $h > 0$ is constant. Residents own wealth W (capital at home or abroad) and save a fraction $s \in (0,1)$ of gross national income (wages plus the gross return on wealth): $\dot{W} = s\,\text{GNI} - \delta W$. Let $N \equiv n + g + \delta$; per effective worker, $k = K/(AhL)$, $y = Y/(AhL)$, $\omega = W/(AhL)$.

a) (3 points) Characterize capital and GDP per effective worker, k^* and y^*, and their dynamics. What does a rise in s do to GDP per worker? Under perfect mobility, which ratio of labor-augmenting productivity A_{BR}/A_{US} (this question's A) reconciles the gaps ϕ and χ of Question 3(a), and what does it say about the Lucas paradox?

Step — capital is pinned (firm's FOC). Firms maximize $K^\alpha(AhL)^{1-\alpha} - (r^* + \delta)K - wL$: $\alpha k^{\alpha-1} = r^* + \delta$, whose left side falls from ∞ to 0. The unique root holds at every date (capital flows in or out instantly):

$$k^* = \Big(\frac{\alpha}{r^* + \delta}\Big)^{\frac{1}{1-\alpha}}, \qquad y^* = k^{*\alpha} = \Big(\frac{\alpha}{r^* + \delta}\Big)^{\frac{\alpha}{1-\alpha}}, \qquad (r^* + \delta)k^* = \alpha y^*.$$

So GDP per worker Ahy^* grows at g from $t = 0$ with no transition, $\partial y^*/\partial s = 0$, and wages per effective worker are $(1 - \alpha)y^*$.

Step — Lucas paradox (link to 3(a)). The same r^*, α, δ give the same k^*, y^*, so output per worker is proportional to Ah: $\phi = (A_{BR}/A_{US})\chi$, i.e. $A_{BR}/A_{US} = \phi/\chi < 1$; in the Hicks-neutral units of 3(a), $(\phi/\chi)^{1-\alpha}$, which sets its MPK ratio to 1.

Economic intuition. Openness makes capital, not saving, adjust; small capital flows reveal a productivity gap.

Ref: Romer (2012), problem 4.4; Kurlat (2020), §5.3.

Grading rubric.

- k^* from the firm's FOC, unique: 1 pt (quoted: max. a third of the item); y^* constant, $\partial y^*/\partial s = 0$: 1 pt.
- $A_{BR}/A_{US} = \phi/\chi$ (or $(\phi/\chi)^{1-\alpha}$ in the Hicks-neutral units of 3(a)): 1 pt. Paradox read without the link to 3(a): max. half of the item.

b) (3 points) Derive the law of motion of wealth per effective worker, its steady state ω^* and the condition for stability. For which world rates is wealth stable at every saving rate?

Step — national income. Wages are $(1 - \alpha)Y$ and wealth earns $r^* + \delta$ gross: $\text{GNI}/(AhL) = (1 - \alpha)y^* + (r^* + \delta)\omega$.

Step — law of motion. $\dot\omega/\omega = \dot W/W - n - g$ (h constant) and the saving rule give

$$\dot\omega = s(1 - \alpha)y^* - \Lambda\,\omega, \qquad \Lambda \equiv N - s(r^* + \delta).$$

Step — solution and stability. Linear: $\omega(t) = \omega^* + (\omega(0) - \omega^*)e^{-\Lambda t}$, $\omega^* = s(1 - \alpha)y^*/\Lambda$, half-life $\ln 2/\Lambda$; it converges iff $\Lambda > 0$, i.e. $s < N/(r^* + \delta)$. If $\Lambda \le 0$, $\dot\omega \ge s(1 - \alpha)y^* > 0$ for $\omega \ge 0$: wealth explodes. As Λ falls with s and is $n + g - r^*$ at $s = 1$, wealth is stable at every $s \in (0, 1)$ iff $r^* \le n + g$.

Economic intuition. A unit of wealth adds $s(r^* + \delta)$ through reinvested income and loses N to depreciation and dilution.

Ref: Romer (2012), §1.3 (same method) and §1.5 (speed of convergence).

Grading rubric.

- GNI per effective worker: 1 pt (return on wealth omitted: max. 1 pt in the item); law of motion and solution: 1 pt.
- $\Lambda > 0$ proved, case $r^* \le n + g$: 1 pt. Stability asserted without the solution: 0 for this line.

c) (3 points) Assume from now on that wealth converges. Find the saving rate $\bar s$ above which Brazil is a net creditor in the steady state and interpret it; compare it with the Golden-Rule saving rate that Question 3(b) gives for this technology. Find GNI/GDP in the steady state in closed form.

Step — net foreign assets. They are $\omega^* - k^*$ per effective worker. By $(r^* + \delta)k^* = \alpha y^*$ from (a), and $s(1 - \alpha)(r^* + \delta) - \alpha\Lambda = s(r^* + \delta) - \alpha N$:

$$\omega^* - k^* = \frac{s(1 - \alpha)y^*}{\Lambda} - \frac{\alpha y^*}{r^* + \delta} = \frac{y^*(s - \bar s)}{\Lambda}, \qquad \bar s \equiv \frac{\alpha N}{r^* + \delta} = \frac{Nk^*}{y^*}.$$

With $\Lambda > 0$, Brazil is a net creditor iff $s > \bar s$.

Step — reading and Golden Rule. $\bar s$ sustains k^* in a closed economy ($\bar s y^* = Nk^*$); $\bar s < N/(r^* + \delta)$ as $\alpha < 1$, and $\bar s < 1$ iff $r^* + \delta > \alpha N$. Cobb–Douglas gives $s_{GR} = \alpha$ (3(b), $\sigma \to 1$), and

$\bar{s} - \alpha = \alpha(n+g-r^*)/(r^*+\delta)$: if $r^* > n+g$ a Golden-Rule saver is a creditor; at $r^* = n+g$, $\bar{s} = \alpha$ and $k^* = k_{GR}$ (as $f'(k^*) = N$).

Step — GNI/GDP. From (b), GNI/GDP $= (1-\alpha) + (r^*+\delta)\omega^*/y^* = (1-\alpha)[1 + s(r^*+\delta)/\Lambda]$, so

$$\frac{\text{GNI}}{\text{GDP}} = \frac{(1-\alpha)N}{\Lambda} = (1-\alpha) + \alpha\frac{\omega^*}{k^*}, \qquad \frac{\omega^*}{k^*} = \frac{s(1-\alpha)(r^*+\delta)}{\alpha\Lambda},$$

and GNI/GDP $- 1 = (r^*+\delta)(s-\bar{s})/\Lambda$: creditor status and GNI above GDP are the same statement.

Economic intuition. Net foreign assets earn the world return: GNI exceeds GDP exactly when residents own more than home capital.

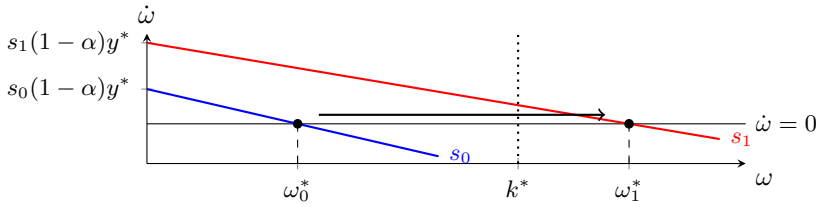
Ref: Romer (2012), §1.4 and problem 4.4; Kurlat (2020), §2.1 (GNP versus GDP).

Grading rubric.

- NFA and creditor condition, using $\Lambda > 0$: 1.5 pts (threshold quoted: max. a third of the item).
- Comparison with s_{GR}: 0.5 pt (one parameter point only: max. half); GNI/GDP and its link to $\bar{s}$: 1 pt.

d) (3 points) At $t = 0$ the saving rate rises once and for all from s_0 to s_1, with $s_0 < \bar{s} < s_1$, both stable. Draw $\dot{\omega}$ against ω for both rates, marking k^* and the steady states. Find the date T at which Brazil becomes a net creditor, and the paths of GNI/GDP and GDP per worker.

Step — phase lines (from (b)). $\dot{\omega} = s_i(1-\alpha)y^* - \Lambda_i\omega$, $\Lambda_i = N - s_i(r^*+\delta)$: intercept $s_i(1-\alpha)y^*$, slope $-\Lambda_i$, zero ω_i^*. As $\partial\dot{\omega}/\partial s = (1-\alpha)y^* + (r^*+\delta)\omega > 0$, the s_1 line is above and flatter ($\Lambda_1 < \Lambda_0$); by (c), $\omega_0^* < k^* < \omega_1^*$.



Step — path and crossing date. From $\omega(0) = \omega_0^*$, $\omega(t) = \omega_1^* - (\omega_1^* - \omega_0^*)e^{-\Lambda_1 t}$, increasing. $\omega(T) = k^*$ and (b)–(c), where y^* cancels, give

$$T = \frac{1}{\Lambda_1}\ln\frac{\omega_1^* - \omega_0^*}{\omega_1^* - k^*} = \frac{1}{\Lambda_1}\ln\frac{(1-\alpha)N(s_1 - s_0)}{\Lambda_0(s_1 - \bar{s})} > 0,$$

as numerator minus denominator is $(\bar{s} - s_0)\Lambda_1 > 0$; $T \to 0$ as $s_0 \to \bar{s}$ and $T \to \infty$ as $s_1 \downarrow \bar{s}$.

Step — GNI and GDP. GNI/GDP $= (1-\alpha) + \alpha\,\omega(t)/k^*$ rises from $(1-\alpha)N/\Lambda_0 < 1$ to $(1-\alpha)N/\Lambda_1 > 1$, crossing 1 at T; the half-life rises to $\ln 2/\Lambda_1$; GDP per worker stays on Ahy^*.

Economic intuition. The reform is wealth accumulation, not capital deepening: GNI moves, GDP does not.

Ref: Romer (2012), §1.4 (a change in the saving rate) and problem 4.4.

Grading rubric.

- Graph: axes, labels, intercepts, slopes, s_1 above and flatter: 0.5 pt; $\omega_0^* < k^* < \omega_1^*$, motion: 0.5 pt.
- $\omega(t)$, T: 1 pt; $T > 0$ proved: 0.5 pt (asserted: 0 for this line); GNI/GDP and GDP: 0.5 pt.

e) (3 points) *Parameter analysis.* How do ω^*/k^*, GNI/GDP, the half-life of wealth and $\bar{s}$ respond to s, r^*, $n+g$, δ and α? Prove each sign, collect the signs in a table, give the limits as $s \to 0$ and as s approaches its upper bound, and read the knife-edge cases $s = \bar{s}$ and $r^* = n+g$.

Step — closed forms (from (b)–(c)). With $R \equiv r^* + \delta$, $\Lambda = N - sR > 0$: $\omega^*/k^* = s(1-\alpha)R/(\alpha\Lambda)$, GNI/GDP $= (1-\alpha) + \alpha\,\omega^*/k^* = (1-\alpha)N/\Lambda$, half-life $\ln 2/\Lambda$, $\bar{s} = \alpha N/R$. The derivatives of Λ in s, r^*, $n+g$, δ, α are $-R$, $-s$, 1, $1-s > 0$, 0.

Step — s, r^*, $n+g$. The half-life moves against Λ: $+,+,-$. Except in α, GNI/GDP moves with ω^*/k^*, whose derivatives are $(1-\alpha)NR/(\alpha\Lambda^2)>0$, $(1-\alpha)Ns/(\alpha\Lambda^2)>0$, $-(1-\alpha)sR/(\alpha\Lambda^2)<0$. For $\bar{s}$: 0, $-\alpha N/R^2<0$, $\alpha/R>0$.

Step — δ and α. δ raises N, R one for one: $\partial(R/\Lambda)/\partial\delta=(N-R)/\Lambda^2$, $\partial\bar{s}/\partial\delta=\alpha(R-N)/R^2$, $N-R=n+g-r^*$: ω^*/k^* and GNI/GDP move with $\operatorname{sgn}(n+g-r^*)$, $\bar{s}$ against it; the half-life falls. α enters only via $(1-\alpha)/\alpha$, $1-\alpha$, α: ω^*/k^* and GNI/GDP fall, $\bar{s}$ rises, the half-life is unchanged.

	s	r^*	$n+g$	δ	α
ω^*/k^*, GNI/GDP	$+$	$+$	$-$	$\operatorname{sgn}(n+g-r^*)$	$-$
half-life	$+$	$+$	$-$	$-$	0
$\bar{s}$	0	$-$	$+$	$\operatorname{sgn}(r^*-n-g)$	$+$

Step — limits and knife-edges. $s\to 0$: $\omega^*/k^*\to 0$, GNI/GDP $\to 1-\alpha$ (only wages), half-life $\to \ln 2/N$. As $s\uparrow\min\{1,N/R\}$: if $r^*\geq n+g$, $\Lambda\downarrow 0$ and all three explode (smallness fails first); if $r^*<n+g$ they stay finite (GNI/GDP $\to(1-\alpha)N/(n+g-r^*)$). At $s=\bar{s}$: $\omega^*=k^*$, GNI = GDP, $\Lambda=(1-\alpha)N$, the closed-economy speed. At $r^*=n+g$: δ drops out of both ratios and $\bar{s}=\alpha=s_{GR}$.

Economic intuition. Openness separates where capital works from who owns it: more return reinvested from wealth (s, r^*) relative to dilution ($n+g$) enriches residents at unchanged output; α shrinks the wage bill that funds saving; δ cuts $k^*/y^*=\alpha/R$ and raises Λ, the first more iff $r^*<n+g$.

Ref: Romer (2012), §1.5 and problem 4.4; Kurlat (2020), §1.1 (GDP) and §2.1 (GNP).

Grading rubric.

- Λ's derivatives and the twenty signs, proved: 1.5 pts (a sign without proof: 0 for that entry).
- Table, limits, knife-edges: 1 pt. Column readings: 0.5 pt (no proofs or no reading: max. half).

Block 3 — Integrative: growth, welfare, national accounts and prices

(Kurlat chs. 1–2, 4–5; Romer ch. 1 • 25 points)

Question 5. (10 points) Objective items.

a) (3 points) *PPP × growth (Hsieh–Klenow).* Brazil (B) and the US (U) produce a final good with $Y=K^{\alpha}(AL)^{1-\alpha}$, $\alpha\in(0,1)$, with the same A and the same $n+g+\delta>0$; both are on their Solow balanced paths. The final good is consumed one for one or turned into new capital at a cost of p_j units of the good per unit of capital, so p_j is the price of investment relative to consumption; $p_B=\pi p_U$ with $\pi>1$, and US prices are the international prices. At domestic prices, investment absorbs the shares $s_B, s_U\in(0,1)$ of output. Output per effective worker, in units of the final good, in Brazil relative to the US is:

A. $\left(s_B/(\pi s_U)\right)^{1/(1-\alpha)}$

B. $\left(s_B/(\pi s_U)\right)^{\alpha/(1-\alpha)}$

C. $\left(s_B/s_U\right)^{\alpha}$

D. $\left(s_B/(\pi s_U)\right)^{\alpha}$

Answer: B

Step — accumulation in physical units. Spending s_jY_j at the price p_j buys $I_j=s_jY_j/p_j$ units of capital, so $\dot{\tilde{k}}=(s_j/p_j)\tilde{k}^{\alpha}-(n+g+\delta)\tilde{k}$, $\tilde{k}_j^*=\left[s_j/(p_j(n+g+\delta))\right]^{1/(1-\alpha)}$ and $\tilde{y}_j^*=(\tilde{k}_j^*)^{\alpha}=\left[s_j/(p_j(n+g+\delta))\right]^{\alpha/(1-\alpha)}$.

Step — ratio. With $p_B=\pi p_U$, $\tilde{y}_B^*/\tilde{y}_U^*=\left[(s_B/p_B)/(s_U/p_U)\right]^{\alpha/(1-\alpha)}=\left(s_B/(\pi s_U)\right)^{\alpha/(1-\alpha)}$, where $s_B/\pi=p_UI_B/Y_B$ is Brazil's investment rate at international prices. Even with $s_B=s_U$ the gap is $\pi^{-\alpha/(1-\alpha)}<1$.

Step — distractors. A is the ratio $\tilde{k}^*_B/\tilde{k}^*_U$ of capital, not of output. D treats $\tilde{k}^*$ as proportional to the real investment rate and misses the feedback from capital to the output that is reinvested (the multiplier $1/(1-\alpha)$). C makes two mistakes: it misses the multiplier, as D does, and measures investment at domestic prices, as if a unit of investment spending bought as much capital in Brazil as in the US.

Economic intuition. Poor countries do not spend a smaller share of output on investment; investment goods are dear there relative to consumption, so the same share buys fewer machines and sustains less capital.

Ref: Kurlat (2020), ch. 1, §1.2 (Making Comparisons); ch. 5, §5.3 (The Capital Accumulation Hypothesis). Hsieh and Klenow (2007), *American Economic Review* 97(3).

b) (3 points) *National accounts × Golden Rule.* An economy with $Y = F(K, AL)$ (constant returns, $\tilde{y} = f(\tilde{k})$, $f' > 0 > f''$, Inada conditions) and competitive factor markets is on a balanced path on which output grows at rate $n + g > 0$. Its national accounts report a gross capital income share $\alpha_K \in (0, 1)$ of GDP and a capital–output ratio $v = K/Y > 0$; capital depreciates at rate $\delta > 0$, and households discount utility at rate $\rho > 0$. Capital per effective worker is strictly below its Golden Rule level, so the path is dynamically efficient, if and only if:

A. $\alpha_K/v > n + g$

B. $\alpha_K/v - \delta > n + g$

C. $\alpha_K/v > \rho + n + g$

D. $\alpha_K/v - \delta > \rho + n + g$

Answer: B

Step — gross return from the accounts. Competitive firms pay capital its marginal product, so gross capital income is $f'(\tilde{k})K$ and $\alpha_K = f'(\tilde{k})\, v$: the accounts reveal $f'(\tilde{k}) = \alpha_K/v$ for any neoclassical f, Cobb–Douglas or not.

Step — Golden Rule. On a balanced path $\hat{c} = f(\tilde{k}) - (n + g + \delta)\tilde{k}$, strictly concave, with its peak at $f'(\tilde{k}_{GR}) = n + g + \delta$. As f' is decreasing, $\tilde{k} < \tilde{k}_{GR} \iff f'(\tilde{k}) > n + g + \delta \iff \alpha_K/v - \delta > n + g$. Multiplying by v: $\alpha_K > (n + g + \delta)v = I/Y$, so gross capital income exceeds the gross investment the balanced path needs; since $I/Y = s$ on that path, the test reads $\alpha_K > s$.

Step — distractors. A forgets depreciation and passes economies above the Golden Rule, with $n+g < \alpha_K/v < n+g+\delta$. D tests capital against the modified golden rule $f' - \delta = \rho + n + g$ of a planner who discounts at ρ (Question 6(d)) and fails efficient economies, with $n+g+\delta < \alpha_K/v < n+g+\delta+\rho$: an optimality condition given preferences, while efficiency is a property of the feasible set, which ρ does not enter. C makes both mistakes: it passes inefficient economies if $\rho < \delta$ and fails efficient ones if $\rho > \delta$.

Economic intuition. Capital that pays out more than it absorbs to keep $\tilde{k}$ constant is a net source of consumption; if it absorbs more, saving less raises consumption at every date (Abel et al., 1989).

Ref: Kurlat (2020), ch. 4, §4.3 (The Golden Rule) and §4.4 (Markets); Romer (2012), ch. 2, §2.11 (The Possibility of Dynamic Inefficiency).

c) (2 points, T or F) *Growth × welfare × discounting.* A Solow economy with $Y = K^\alpha (AL)^{1-\alpha}$, $\alpha \in (0, 1)$, $\dot{A}/A = g$, $\dot{L}/L = n$ and $n + g + \delta > 0$ is on its balanced path with a saving rate s_0 below the Golden Rule rate s_{GR}. Current households value the path of consumption per worker by $W = \int_0^\infty e^{-\rho t} \ln c(t)\, dt$. A permanent rise in the saving rate at $t = 0$, to any $s_1 \in (s_0, s_{GR}]$, raises W for every discount rate $\rho > 0$.

Answer: F

Step — integration by parts. $A(t)$ is common to both paths, so $\Delta W = \int_0^\infty e^{-\rho t} D(t)\, dt$, where $D(t) = \ln \hat{c}_1(t) - \ln \hat{c}^*_0$ is the bounded log change in consumption. Integrating by parts,

$$\rho\, \Delta W = D(0) + \int_0^\infty e^{-\rho t} D'(t)\, dt.$$

Step — impact and transition. $\tilde{k}$ is predetermined, so $D(0) = \ln[(1-s_1)/(1-s_0)] < 0$. Afterwards $D'(t) = \alpha\dot{\tilde{k}}/\tilde{k} = \alpha[s_1\tilde{k}^{\alpha-1} - (n+g+\delta)]$ is positive and falls as $\tilde{k}$ rises, so $0 < D'(t) \leq D'(0) = \alpha(n+g+\delta)(s_1-s_0)/s_0$.

Step — the sign depends on ρ. Hence $D(0) < \rho\,\Delta W < D(0) + D'(0)/\rho$, and $\rho\,\Delta W \to D(0) < 0$ as $\rho \to \infty$; explicitly, $\Delta W < 0$ for every $\rho \geq \bar{\rho} \equiv D'(0)/|D(0)|$, that is,

$$\bar{\rho} = \frac{\alpha(n+g+\delta)(s_1-s_0)}{s_0 \ln[(1-s_0)/(1-s_1)]}.$$

As $\rho \to 0$, instead, monotone convergence gives $\rho\,\Delta W \to D(\infty) = \ln\frac{1-s_1}{1-s_0} + \frac{\alpha}{1-\alpha}\ln\frac{s_1}{s_0} > 0$, because $\frac{d}{ds}\ln\hat{c}^*(s) = \frac{\alpha-s}{(1-\alpha)s(1-s)} > 0$ below $s_{GR} = \alpha$. The sign of ΔW depends on ρ; Question 6 develops the threshold.

Economic intuition. Saving more trades an immediate consumption cut for a gain that builds up with capital; being below the Golden Rule guarantees the gain, not that it is worth the wait.

Ref: Kurlat (2020), ch. 2, §2.2 (Beyond GDP); ch. 4, §4.3 (The Golden Rule); Romer (2012), ch. 1, §1.4 (The Impact of a Change in the Saving Rate).

d) (2 points, T or F) *Measurement × growth.* In the economy of item c), let the saving rate rise permanently at $t = 0$ from s_0 to any $s_1 \in (s_0, 1)$. Output per worker does not jump but consumption per worker falls; afterwards both grow faster than g, at the same rate, until the new balanced path.
Answer: T

Step — impact. K, L and A are predetermined, so $y = A\tilde{k}^\alpha$ does not jump, while $c = (1-s)y$ falls by the factor $(1-s_1)/(1-s_0) < 1$.

Step — transition. With s_1 fixed, $c/y = 1 - s_1$, so $\dot{c}/c = \dot{y}/y = g + \alpha\dot{\tilde{k}}/\tilde{k}$, where $\dot{\tilde{k}}/\tilde{k} = s_1\tilde{k}^{\alpha-1} - (n+g+\delta) > 0$ for $\tilde{k} \in [\tilde{k}_0^*, \tilde{k}_1^*)$. The excess over g falls monotonically to zero as $\tilde{k} \to \tilde{k}_1^*$, which is reached only asymptotically.

Step — what the accounts show. Growth accounting with $k = K/L$ gives $\dot{y}/y = \alpha\dot{k}/k + (1-\alpha)g$: the residual stays at $(1-\alpha)g$, so the whole acceleration is capital deepening, and the consumption share of GDP drops once, from $1 - s_0$ to $1 - s_1$.

Ref: Kurlat (2020), ch. 4, §4.5 (Technological Progress); ch. 5, §5.4 (Growth Accounting); Romer (2012), ch. 1, §1.4 and §1.7.

Question 6. (15 points) Discursive. *Integrative: Solow transition × national accounts × welfare measurement × discounting.* Brazil debates a pension reform that would raise national saving for good. Take a closed Solow economy without government: $Y = K^\alpha(AL)^{1-\alpha}$, $0 < \alpha < 1$; A grows at rate g, L at rate n, capital depreciates at rate δ, and $N \equiv n + g + \delta > 0$. Per effective worker, $\tilde{y} = \tilde{k}^\alpha$ and $\hat{c} = (1-s)\tilde{y}$. The economy is on its balanced growth path with saving rate s_0; at $t = 0$ the reform raises the saving rate permanently to s_1, with $0 < s_0 < s_1 \leq \alpha$. A representative household values the path of consumption per person, $c(t) = A(t)\hat{c}(t)$, by $W = \int_0^\infty e^{-\rho t}\ln c(t)\,dt$, $\rho > 0$. Let $D(t) \equiv \ln\hat{c}_1(t) - \ln\hat{c}_0$ be the log gap between consumption per effective worker with the reform and without it ($\hat{c}_0$ is constant).

a) (3 points) Using the change of variable $z \equiv \tilde{k}^{1-\alpha}$, find the exact path of $\tilde{k}$ after the reform and its speed of convergence λ. Derive $D(t)$, its impact value $D(0)$ and its long-run value $D(\infty)$, and prove that $D(\infty) > 0$. Find the recovery time T_R at which consumption regains its pre-reform level, and the condition under which it exists.

Step — law of motion. From $\dot{K} = sY - \delta K$, with AL growing at $n + g$: $\dot{\tilde{k}} = s_1\tilde{k}^\alpha - N\tilde{k}$ for $t \geq 0$, from the predetermined $\tilde{k}(0) = \tilde{k}_0^* = (s_0/N)^{1/(1-\alpha)}$. With $z = \tilde{k}^{1-\alpha}$ this Bernoulli equation becomes linear:

$$\dot{z} = (1-\alpha)\tilde{k}^{-\alpha}\dot{\tilde{k}} = (1-\alpha)(s_1 - Nz), \qquad z(0) = s_0/N.$$

Step — exact path. Hence

$$z(t) = \frac{s_1 - (s_1 - s_0)e^{-\lambda t}}{N}, \qquad \lambda \equiv (1-\alpha)N, \qquad \tilde{k}(t) = z(t)^{1/(1-\alpha)},$$

exact at any distance from the new steady state: the gap $s_1/N - z$ closes at the rate λ, which does not depend on s.

Step — consumption gap. $\tilde{y} = z^{\zeta}$ with $\zeta \equiv \frac{\alpha}{1-\alpha}$, so $\hat{c}_1(t) = (1-s_1)z(t)^{\zeta}$ and $\hat{c}_0 = (1-s_0)(s_0/N)^{\zeta}$. With $b \equiv \frac{s_1-s_0}{s_1} \in (0,1)$:

$$D(t) = \ln\frac{1-s_1}{1-s_0} + \zeta \ln\Big[1 + \frac{s_1-s_0}{s_0}\big(1-e^{-\lambda t}\big)\Big] = D(\infty) + \zeta \ln\big(1 - b\,e^{-\lambda t}\big),$$

$$D(0) = \ln\frac{1-s_1}{1-s_0} < 0, \qquad D(\infty) = D(0) + \zeta \ln\frac{s_1}{s_0}, \qquad D'(t) = \frac{\alpha N\, b\, e^{-\lambda t}}{1 - b\, e^{-\lambda t}} > 0.$$

Step — long-run gain. $D(\infty) = G(s_1) - G(s_0)$ with $G(s) \equiv \ln(1-s) + \zeta \ln s$, the log of steady-state $\hat{c}$ up to a constant. $G'(s) = \frac{\alpha - s}{(1-\alpha)s(1-s)}$ is positive on $(0,\alpha)$ and zero at α, so G is strictly increasing on $(0,\alpha]$ and $D(\infty) > 0$ whenever $s_0 < s_1 \le \alpha$.

Step — recovery time. D is continuous and strictly increasing from $D(0) < 0$ toward $D(\infty)$. A root exists iff $D(\infty) > 0$ (intermediate value theorem; if $D(\infty) \le 0$, then $D(t) < D(\infty) \le 0$ at every t), and it is unique. Solving $D(T_R) = 0$ with $R \equiv \big(\frac{1-s_0}{1-s_1}\big)^{(1-\alpha)/\alpha} = e^{-D(0)/\zeta}$:

$$T_R = \frac{1}{\lambda}\ln\frac{s_1 - s_0}{s_1 - R\,s_0} = -\frac{1}{\lambda}\ln\Big[1 - \frac{s_0(R-1)}{s_1 - s_0}\Big],$$

well defined iff $s_1 > Rs_0$, which after taking logs is $\zeta \ln(s_1/s_0) > -D(0)$, i.e. $D(\infty) > 0$.

Economic intuition. The reform is an investment by the whole economy: consumption falls on impact by the extra share of output saved, and the extra capital pays back gradually, at the speed λ of the Solow transition.

Ref: Kurlat (2020), §4.2 and §4.5; Romer (2012), §1.4 and §1.5 (the linearized speed there is exact here for z).

Grading rubric.

- Law of motion, linear equation in z, exact path and λ: 1 pt (linearized path used as exact: max. half of this line).
- $D(t)$, $D(0)$, $D(\infty)$ and the proof that $D(\infty) > 0$: 1 pt ($D(\infty) > 0$ asserted without proof: 0 for that part).
- T_R with its existence condition, proved by monotonicity: 1 pt.

b) (3 points) Describe the consumption and investment shares of GDP and the growth rates of GDP and of consumption per worker along the transition. What does GDP growth say, and fail to say, about the household's welfare? Draw $D(t)$ and $\ln\tilde{y}_1(t) - \ln\tilde{y}_0$ against t, marking $D(0)$, T_R and the long-run values.

Step — shares. Closed economy without government: $Y = C + I$ with $I = sY$, so $C/Y = 1 - s$ and $I/Y = s$ jump at $t = 0$ (from $1 - s_0$ and s_0 to $1 - s_1$ and s_1) and then stay constant. GDP does not jump, because $\tilde{k}(0)$ is predetermined: the reform changes the composition of expenditure, not its level.

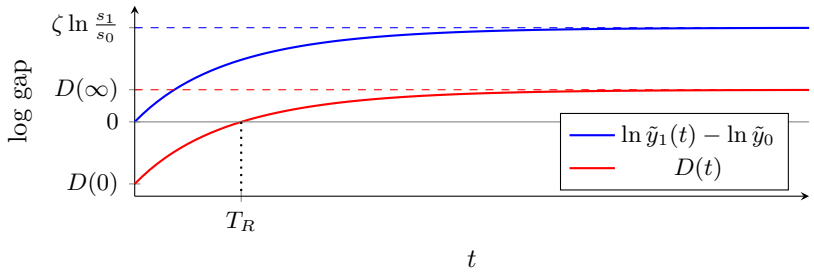
Step — growth rates. For $t > 0$, $y = Y/L = A\tilde{y}$ and $c = (1-s_1)A\tilde{y}$ both grow at

$$\gamma(t) = g + \zeta\,\frac{\dot{z}}{z} = g + \frac{\alpha N(s_1 - s_0)e^{-\lambda t}}{s_1 - (s_1 - s_0)e^{-\lambda t}},$$

which exceeds g, falls strictly (the numerator falls, the positive denominator rises) and tends to g, starting from $g + \alpha N(s_1 - s_0)/s_0$. Growth accounting attributes the excess to capital deepening: TFP growth $(1-\alpha)g$ is unchanged and $\alpha\,\dot{\tilde{k}}/\tilde{k} = \zeta\,\dot{z}/z$.

Step — output versus consumption. The output gap $\ln\tilde{y}_1(t) - \ln\tilde{y}_0 = \zeta \ln\big[1 + \frac{s_1-s_0}{s_0}(1-e^{-\lambda t})\big]$ rises from 0 to $\zeta\ln(s_1/s_0)$; $D(t)$ is the same curve shifted down by $-D(0) = \ln\frac{1-s_0}{1-s_1}$, the extra share of output invested.

Step — what GDP growth says. GDP per worker is higher at every $t > 0$ and grows faster than g, so GDP growth records the benefits and none of the cost: the level fall of consumption at $t = 0$ leaves GDP unchanged and shows up only in C/Y. As welfare is $\int_0^\infty e^{-\rho t} D\, dt$, the same GDP path fits a gain or a loss, depending on ρ (part c).



Economic intuition. GDP measures what is produced, not what is consumed: saving more moves output from consumption to investment without changing GDP, so the sacrifice shows up only in the composition of GDP.

Ref: Kurlat (2020), §1.1 (expenditure approach), §2.2 (beyond GDP) and §5.4 (growth accounting).

Grading rubric.

- Shares of GDP, their jump at $t = 0$, GDP continuous: 0.5 pt.
- Growth rate in closed form, proved to exceed g, fall and tend to g: 1 pt.
- Welfare reading (benefits counted, the level cost not; the sign depends on ρ): 0.5 pt. No link from the national accounts to welfare: max. half of the item.
- Graph: axes, the two curves as vertical translates from 0 and $D(0)$, with their asymptotes: 0.5 pt; $D(0)$, T_R and $D(\infty)$ marked: 0.5 pt.

c) (3 points) Let $\Delta W(\rho)$ be the welfare gain of the reform and $x(\rho)$ its consumption equivalent: the permanent proportional rise in the no-reform consumption path that the household values as much as the reform. Find x in terms of ΔW, show that x is strictly decreasing in ρ, find its limits as $\rho \to 0$ and $\rho \to \infty$, and prove that there is exactly one welfare-neutral discount rate ρ^*. How does ρ^* depend on N?

Step — welfare gain and consumption equivalent. $\ln c(t) = \ln A(0) + gt + \ln \hat{c}(t)$ and $\ln A(0) + gt$ is common to both paths, so $\Delta W = \int_0^\infty e^{-\rho t} D(t)\, dt$. Scaling the no-reform path by $1 + x$ adds $\int_0^\infty e^{-\rho t} \ln(1+x)\, dt = \ln(1+x)/\rho$ to W_0; equating it to ΔW gives $x(\rho) = e^{V(\rho)} - 1$ with $V \equiv \rho\, \Delta W$.

Step — integration by parts. D is bounded and C^1, so integrating by parts and then expanding D' from (a) as a geometric series ($0 < b\, e^{-\lambda t} < 1$), integrated term by term (all terms positive):

$$V(\rho) = D(0) + \int_0^\infty e^{-\rho t} D'(t)\, dt = D(0) + \alpha N \sum_{k \geq 1} \frac{b^k}{\rho + k\lambda} = D(0) + \zeta \sum_{k \geq 1} \frac{b^k}{\rho/\lambda + k}.$$

Step — monotonicity and limits. Each term falls strictly with ρ, so V and x are strictly decreasing. As $\rho \to \infty$ each term tends to 0 and is bounded by b^k/k (dominated convergence): $V \to D(0)$ and $x \to e^{D(0)} - 1 = -\frac{s_1 - s_0}{1 - s_0}$. As $\rho \to 0$ the terms rise to b^k/k, and $\sum_k b^k/k = -\ln(1-b) = \ln \frac{s_1}{s_0}$ (monotone convergence): $V \to D(\infty)$ and $x \to e^{D(\infty)} - 1 > 0$, the steady-state gain $\hat{c}^*(s_1)/\hat{c}^*(s_0) - 1$.

Step — welfare-neutral rate. V is continuous and strictly decreasing from $D(\infty) > 0$ to $D(0) < 0$. By the intermediate value theorem there is ρ^* with $x(\rho^*) = 0$, unique by strict monotonicity; the reform raises welfare iff $\rho < \rho^*$.

Step — scaling in N. V depends on ρ only through ρ/λ, and otherwise only on (s_0, s_1, α), so $\rho^* = (1 - \alpha) N \varphi(s_0, s_1, \alpha)$ is proportional to N: D depends on t only through λt, and a faster transition just changes the time unit.

Economic intuition. A patient household ($\rho \to 0$) counts only the new steady state, an impatient one ($\rho \to \infty$) only the impact loss; ρ^* balances the two, in units of the transition speed λ.

Ref: Kurlat (2020), §2.2 (consumption-equivalent welfare) and §6.3 (discounting over many periods).

Grading rubric.

- ΔW (the common trend cancels) and $x = e^{\rho \Delta W} - 1$: 0.75 pt.
- Integration by parts (or the series) and strict monotonicity, proved: 1 pt.
- Both limits, with their reading: 0.5 pt (a limit stated without proof: 0 for it).
- Existence and uniqueness of ρ^* (intermediate value theorem plus monotonicity): 0.5 pt; $\rho^* \propto N$: 0.25 pt.

d) (3 points) For a marginal reform ($s_1 \downarrow s_0$), find the sign of the derivative of $\rho\,\Delta W$ with respect to s_1 and the discount rate ρ_m at which it changes sign. Relate ρ_m to the modified golden rule of a planner who maximizes W, and prove that $\rho^* < \rho_m$ for every reform with $s_1 > s_0$.

Step — linearized dynamics. Write $s_1 = s_0 + ds$ and linearize $\dot{\tilde{k}} = s_1 \tilde{k}^\alpha - N\tilde{k}$ at $\tilde{k}_0^*$, where $s_0 \alpha (\tilde{k}_0^*)^{\alpha-1} = \alpha N$:

$$\frac{d\,\Delta\tilde{k}}{dt} = \tilde{y}_0^*\,ds - (1-\alpha)N\,\Delta\tilde{k}, \quad \Delta\tilde{k}(0) = 0 \implies \Delta\tilde{k}(t) = \frac{\tilde{y}_0^*}{\lambda}(1 - e^{-\lambda t})\,ds,$$

which is also $\partial\tilde{k}(t)/\partial s_1$ of the exact path of (a) at $s_1 = s_0$.

Step — marginal welfare. From $\hat{c} = (1-s)\tilde{k}^\alpha$, $\Delta \ln \hat{c} = -\frac{ds}{1-s_0} + \alpha\frac{\Delta\tilde{k}}{\tilde{k}_0^*}$, and $\alpha\tilde{y}_0^*/\tilde{k}_0^* = f'(\tilde{k}_0^*) = \alpha N/s_0$. With $\rho\int_0^\infty e^{-\rho t}(1-e^{-\lambda t})\,dt = \frac{\lambda}{\rho+\lambda}$:

$$\frac{\partial V}{\partial s_1}\Big|_{s_1=s_0} = -\frac{1}{1-s_0} + \frac{f'(\tilde{k}_0^*)}{\rho+\lambda} > 0 \iff \rho < (1-s_0)f'(\tilde{k}_0^*) - \lambda = f'(\tilde{k}_0^*) - N \equiv \rho_m,$$

where the last equality uses $s_0 f'(\tilde{k}_0^*) = \alpha N$. So $\rho_m = N(\alpha - s_0)/s_0 > 0$, because $s_0 < \alpha$.

Step — planner. A planner who maximizes W subject to $\dot{\tilde{k}} = f(\tilde{k}) - \hat{c} - N\tilde{k}$ cannot gain by consuming one unit less of $\hat{c}$ at t and the proceeds $1 + (f'(\tilde{k}) - N)\,dt$ at $t + dt$: $1/\hat{c}(t) = e^{-\rho\,dt}\,[1 + (f'(\tilde{k}) - N)\,dt]/\hat{c}(t+dt)$, i.e. the Euler equation $\dot{\hat{c}}/\hat{c} = f'(\tilde{k}) - N - \rho$. Its steady state is the modified golden rule $f'(\tilde{k}) = \rho + N$, i.e. $f' - \delta = \rho + n + g$. So a small reform helps iff $\tilde{k}_0^*$ is below the modified-golden-rule capital of the household's own ρ; at ρ_m it is exactly there.

Step — concavity in s_1. With $E \equiv e^{-\lambda t}$, $D = \ln(1-s_1) + \zeta \ln\big(s_1(1-E) + s_0 E\big) - \ln(1-s_0) - \zeta \ln s_0$ and $\frac{\partial^2 D}{\partial s_1^2} = -\frac{1}{(1-s_1)^2} - \frac{\zeta(1-E)^2}{(s_1(1-E)+s_0E)^2} < 0$ at every t. So $V(s_1;\rho) = \rho\int_0^\infty e^{-\rho t} D\,dt$ is strictly concave in s_1, and $V(s_0;\rho) = 0$.

Step — $\rho^* < \rho_m$. At ρ_m, $V(s_0;\rho_m) = 0$ and $V_{s_1}(s_0;\rho_m) = 0$, so strict concavity gives $V(s_1;\rho_m) < 0 = V(s_1;\rho^*)$ for every $s_1 > s_0$. Since V is strictly decreasing in ρ (part c), $\rho^* < \rho_m$.

Economic intuition. At the margin the reform costs $1/(1-s_0)$ now and earns $f'(\tilde{k}_0^*)$ on capital that builds up at rate λ, worth $f'/(\rho+\lambda)$. Larger reforms meet diminishing returns (concavity in s_1), so they need more patience.

Ref: Kurlat (2020), §4.3 (golden rule) and §6.3 (Euler equation); Romer (2012), §2.5 (modified golden rule).

Grading rubric.

- Linearization and the marginal welfare effect in closed form: 1 pt.
- $\rho_m = f'(\tilde{k}_0^*) - N$ and the modified golden rule derived from the planner's Euler equation: 1 pt (rule quoted, not derived: max. a third of this line).
- Proof of $\rho^* < \rho_m$ (concavity in s_1, $V_{s_1}(s_0;\rho_m) = 0$, monotonicity in ρ): 1 pt (shown for a special case only: max. half of this line).

e) (3 points) *Parameter analysis.* Sign, with proofs, the responses of $D(0)$, $D(\infty)$, T_R, ρ^* and ρ_m to s_1, s_0 and N, and give the limits as $s_1 \downarrow s_0$, at $s_1 = \alpha$, and as $N \to 0$ or $N \to \infty$. Compare the saving rate that maximizes $D(\infty)$ with the one that maximizes the welfare change, and explain the difference. Draw $x(\rho)$ for two reforms $s_1 < s_1'$, marking where each crosses zero.

Step — closed forms. $\partial D(0)/\partial s_1 = -1/(1-s_1) < 0$ and $\partial D(0)/\partial s_0 = 1/(1-s_0) > 0$. $\partial D(\infty)/\partial s_1 = G'(s_1) > 0$ on (s_0, α) and $= 0$ at α, where $D(\infty)$ peaks; $\partial D(\infty)/\partial s_0 = -G'(s_0) < 0$. $\rho_m = N(\alpha - s_0)/s_0$ does not depend on s_1 and falls with s_0.

Step — convexity in s_0. $\frac{\partial D}{\partial s_0} = \frac{1}{1-s_0} - \frac{\zeta s_1(1-E)}{s_0 s_1(1-E)+s_0^2 E}$, whose last denominator rises with s_0, so $\frac{\partial D}{\partial s_0}$ rises: D is strictly convex in s_0 at every t, and so is V. Also $D = V = 0$ when $s_1 = s_0$.

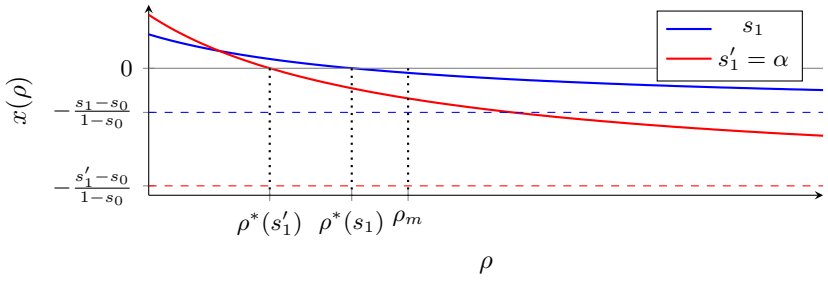
Step — T_R and ρ^* in s_1 and s_0. Fix $t = T_R$. On $[s_0, s_1]$, D vanishes at both ends both as a function of the new rate (strictly concave) and as a function of the old rate (strictly convex), so $D_{s_1} < 0$ at s_1 and $D_{s_0} < 0$ at s_0. With $D_t > 0$, $dT_R/ds_i = -D_{s_i}/D_t > 0$ for $i = 0, 1$ (for s_1, equivalently: $\frac{R-1}{s_1-s_0}$ is the slope of a secant of the convex $R(s_1)$ from $R(s_0) = 1$, so it rises with s_1). The same argument on $V(\cdot\,; \rho^*)$, with $V_\rho < 0$, gives $d\rho^*/ds_i = -V_{s_i}/V_\rho < 0$.

Step — N. By (a) and (c), N enters only through the time scale λt: $D(0)$ and $D(\infty)$ do not depend on N, $T_R \propto 1/N$, and ρ^* and ρ_m are proportional to N.

	$s_1 \uparrow$	$s_0 \uparrow$	$N \uparrow$
$D(0)$ (impact loss)	$-$	$+$	0
$D(\infty)$ (long-run gain)	$+$ on (s_0, α), 0 at α	$-$	0
T_R (recovery time)	$+$	$+$	$-$ $(\propto 1/N)$
ρ^* (welfare-neutral rate)	$-$	$-$	$+$ $(\propto N)$
ρ_m (marginal threshold)	0	$-$	$+$ $(\propto N)$

Step — limits. As $s_1 \downarrow s_0$: $D(0), D(\infty) \to 0$ but $\lambda T_R \to \ln \frac{\alpha(1-s_0)}{\alpha - s_0} > 0$ (l'Hôpital on the first form of T_R): even a tiny reform takes a finite time to pay back. Also $\rho^* \uparrow \rho_m$: for any $\rho < \rho_m$, $V_{s_1}(s_0; \rho) > 0$, so $V(s_1; \rho) > 0$, i.e. $\rho^* > \rho$, for s_1 close to s_0. At $s_1 = \alpha$, $D(\infty)$ and T_R are largest and ρ^* smallest over $(s_0, \alpha]$. As $s_0 \uparrow \alpha$, $\rho_m \to 0$. As $N \to 0$, $T_R \to \infty$ and $\rho^*, \rho_m \to 0$; as $N \to \infty$, $T_R \to 0$ and $\rho^*, \rho_m \to \infty$.

Step — golden rule versus welfare. At $s_1 = \alpha$, $\partial D/\partial s_1 = -s_0 E/\{(1-\alpha)[\alpha(1-E) + s_0 E]\} < 0$ at every t, so $V_{s_1}(\alpha; \rho) < 0$ for every $\rho > 0$: a rate slightly below α gives more welfare. As V is strictly concave in s_1 and $V_{s_1}(s_0; \rho) > 0$ for $\rho < \rho_m$, the best constant rate is unique and lies in (s_0, α); for $\rho \geq \rho_m$ it is s_0 (no reform). At α, $G'(\alpha) = 0$: more saving no longer raises steady-state consumption but still lowers it at every finite date, a cost that stops counting only as $\rho \to 0$, when $V \to D(\infty)$.



Economic intuition. A more ambitious reform (higher s_1) or one that starts with more capital (higher s_0) earns a lower marginal product, so it pays back later and needs more patience; N only changes the clock. The golden-rule reform has the largest long-run gain and the smallest ρ^*.

Ref: Kurlat (2020), §4.3 (golden rule, dynamic efficiency); Romer (2012), §1.4 and §2.5.

Grading rubric.

- Signs in s_1 and s_0, each proved (closed forms; concavity and convexity at T_R and at ρ^*): 1.25 pt.
- Signs in N from the time scaling, and the table: 0.5 pt. Limits, including $\rho^* \to \rho_m$ and the finite T_R of a marginal reform: 0.5 pt. Golden rule versus welfare, with the proof that $V_{s_1}(\alpha; \rho) < 0$: 0.25 pt.
- Graph: two decreasing curves with the limits of (c), crossing zero in the order $\rho^*(s_1') < \rho^*(s_1) < \rho_m$: 0.5 pt. A sign table without proofs or without the economic reading: max. half of the item.